

Week 3 · Day 5 · VPC Peering & Connectivity

Draw Your Complete 3-Tier Architecture Diagram

 Time

55 min

 Objective

Create a professional, complete AWS 3-tier architecture diagram incorporating everything from Week 3: VPC, subnets, route tables, IGW, Security Groups, and EC2 placement.

 Prerequisites — Complete Before Starting

All Week 3 labs completed · Excalidraw or draw.io (app.diagrams.net — free)

 Portfolio Goal — What You Will Post

This diagram is your Week 3 portfolio centrepiece — a professional architecture diagram is the single most impressive artifact you can post on LinkedIn or put in a GitHub repo.

Lab Steps

Step 1 Choose Your Diagramming Tool

1. Professional AWS diagrams use AWS architecture icons. Download the AWS icon set: <https://aws.amazon.com/architecture/icons/>
2. Open draw.io (app.diagrams.net) — it has built-in AWS icon support.
3. Or use Excalidraw for a hand-drawn / whiteboard aesthetic — also professional.
4. Both are free and widely recognised.

Step 2 Layer 1 — VPC & Availability Zones

1. Draw the outer VPC boundary box. Label: OluTech-Production-VPC (10.0.0.0/16).
2. Inside, draw two vertical columns labelled: Availability Zone A and Availability Zone B.
3. In each column, draw two horizontal boxes: Public Subnet (top) and Private Subnet (bottom).
4. Label each subnet with its CIDR range.

Step 3 Layer 2 — Internet Connectivity

1. Outside the VPC box, draw: Internet (cloud icon) → Internet Gateway → connects to Public Subnets.
2. Label the Internet Gateway: OluTech-IGW.

3. Show the route table entry: 0.0.0.0/0 → IGW as a small label on the arrow.
4. Add a NAT Gateway icon in Public Subnet AZ-A with an arrow to Private Subnet AZ-A.

Step 4 Layer 3 — Compute & Services

1. In the Public Subnets: place an Application Load Balancer icon spanning both AZs.
2. In the Private Subnets (upper): place EC2 instance icons labelled 'App Server'.
3. In the Private Subnets (lower): place RDS DB icons labelled 'MySQL RDS'.
4. Show the security group flow with coloured dashed arrows: SG-LB → SG-WebServers → SG-Database.

Step 5 Polish & Export

1. Add a title at the top: 'OluTech Solutions — 3-Tier AWS Architecture'.
2. Add a legend showing what each shape/colour means.
3. Add your name and the date.
4. Export as PNG at high resolution.
5. Also export as PDF for your portfolio.
6. Upload to your GitHub portfolio repo. Commit with message: 'Add Week 3 VPC architecture diagram'.



Screenshot Checklist

Take these screenshots and save them to your portfolio folder:

- ☐ Completed 3-tier architecture diagram (high resolution PNG)
- ☐ GitHub commit showing the diagram added to your portfolio repo



Portfolio Post — Copy & Customise

LinkedIn post featuring the diagram: 'Week 3 complete — built an entire AWS VPC networking stack from scratch and documented it in this architecture diagram. VPC, subnets, route tables, IGW, NAT Gateway, Security Groups, bastion host, 3-tier compute. Built every component hands-on. #AWSarchitecture #CloudPortfolio #SAA-C03 #Week3Done'



Bonus Challenge — Push Your Skills Further

Research VPC Peering and VPC Transit Gateway. Add a second VPC (representing a 'Development' environment) to your diagram and draw a peering connection between Production and Development. Label the limitations of peering (no transitive routing).



Lab Complete — Mark Off Your Progress

Lab Steps Completed	☐
All Screenshots Taken	☐
Portfolio Artifact Posted	☐
Bonus Challenge Attempted	☐